case Greek sigma. Likewise, the imaginary axis uses the variable, ω, the natural frequency. This coordinate system allows the location of any point to be specified by providing values for σ and ω. Using complex notation, each location is represented by the complex variable, *s*, where: $s = \sigma + j\omega$. Just as with the Fourier transform, signals in the s-domain are represented by capital letters. For example, a time domain signal, $x(t)$, is transformed into an s-domain signal, $X(s)$, or alternatively, $X(\sigma,\omega)$. The s-plane is continuous, and extends to infinity in all four directions.

In addition to having a *location* defined by a complex number, each point in the s-domain has a *value* that is a complex number. In other words, each location in the s-plane has a real part and an imaginary part. As with all complex numbers, the real & imaginary parts can alternatively be expressed as the magnitude & phase.

Just as the Fourier transform analyzes signals in terms of sinusoids, the Laplace transform analyzes signals in terms of sinusoids *and* exponentials. From a mathematical standpoint, this makes the Fourier transform a *subset* of the more elaborate Laplace transform. Figure 32-1 shows a graphical description of how the s-domain is related to the time domain. To find the values along a vertical line in the s-plane (the values at a particular σ), the time domain signal is first multiplied by the exponential curve: $e^{-\sigma t}$. The left half of the s-plane multiplies the time domain with exponentials that *increase* with time ($\sigma < 0$), while in the right half the exponentials *decrease* with time ($\sigma > 0$). Next, take the complex Fourier transform of the exponentially weighted signal. The resulting spectrum is placed along a vertical line in the s-plane, with the top half of the s-plane containing the positive frequencies and the bottom half containing the negative frequencies. Take special note that the values on the y-axis of the s-plane ($\sigma = 0$) are exactly equal to the Fourier transform of the time domain signal.

As discussed in the last chapter, the complex Fourier Transform is given by:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

This can be expanded into the Laplace transform by first multiplying the time domain signal by the exponential term:

$$X(\sigma,\omega) = \int_{-\infty}^{\infty} [\, x(t)\, e^{-\sigma t}\,]\, e^{-j\omega t}\, dt$$

While this is not the simplest form of the Laplace transform, it is probably the best description of the strategy and operation of the technique. To